Lu Zhang, Ph.D.

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张鲁 · Battery scientist & data engineer

Battery scientist with 14+ years at Argonne National Laboratory developing redox flow batteries, lithium-ion electrolytes, and silicon-anode binders; now applying machine learning to industrial-scale energy storage. R&D 100 Award (2014) for redox-shuttle technology. Built cross-institution Sprint teams under JCESR (DOE BES).

112 peer-reviewed papers **28** US patents & applications **13** invited talks **3,210** citations h-index **39**

24 first / corresp.

Education

Ph.D., Organic Chemistry — **Chinese Academy of Sciences** (Technical Institute of Physics and Chemistry) 2001 – 2006

Advisor: Prof. 李嫕 (Yi Li). Combined master/PhD program; photochemistry / organic photophysics group.

B.S., Chemistry — **Shandong University** 1996 – 2000

Professional Experience

Data Science Engineer — **American Academy of Orthopaedic Surgeons (AAOS)** Feb 2026 – present

Voice agents and clinical data systems.

Senior Data Scientist — **Mercury Insurance** Nov 2024 – Jan 2026

Pricing and risk models; production ML pipelines.

Data Scientist — **Cincinnati Insurance** Aug 2023 – Nov 2024

Loss prediction and analytics platforms.

Chemist → Senior Chemist — **Argonne National Laboratory** Jan 2009 – Aug 2023

Total tenure 14 years 7 months. JCESR Theme Lead, Sprint team leader, Y3 milestone lead. R&D 100 Award (2014).

Postdoctoral Researcher — **Florida State University** 2006 – 2008

Department of Chemistry and Biochemistry. PI: Prof. Lei Zhu.

Awards & Recognition

- **R&D 100 Award (2014)** — for redox-shuttle additive technology (the "Oscars of Innovation"). Core inventor on lithium-battery overcharge-protection redox-shuttle patents.
- **R&D 100 Award Finalist (2015)** — for high-voltage lithium-battery fluorinated electrolyte technology.
- Featured in Argonne Today, JCESR Research Highlights, EurekAlert, Tech Xplore, PV Magazine, ScienceDaily.

Selected Publications (top 10 by citations)

Full list of 112 peer-reviewed papers + book chapters at rockyzl.github.io/lu-zhang-site/publications.

Radical Compatibility with Nonaqueous Electrolytes and Its Impact on an All-Organic Redox Flow Battery — Angewandte Chemie International Edition, 2015; 322 citations.

A High-Current, Stable Nonaqueous Organic Redox Flow Battery — ACS Energy Letters, 2016; 237 citations.

A symmetric organic-based nonaqueous redox flow battery and its state of charge diagnostics by FTIR — Journal of Materials Chemistry A, 2016; 198 citations.

Molecular engineering towards safer lithium-ion batteries: a highly stable and compatible redox shuttle for overcharge protection — Energy & Environmental Science, 2012; 126 citations. [first author]

Re-Engineering Poly(Acrylic Acid) Binder toward Optimized Electrochemical Performance for Silicon Lithium-Ion Batteries: Branching Architecture Leads to Balanced Properties of Polymeric Binders — Advanced Functional Materials, 2019; 114 citations.

2,5-Dimethoxy-1,4-Benzoquinone (DMBQ) as Organic Cathode for Rechargeable Magnesium-Ion Batteries — Journal of The Electrochemical Society, 2016; 108 citations.

The existence of optimal molecular weight for poly(acrylic acid) binders in silicon/graphite composite anode for lithium-ion batteries — Journal of Power Sources, 2018; 104 citations. [corresponding]

A subtractive approach to molecular engineering of dimethoxybenzene-based redox materials for non-aqueous flow batteries — Journal of Materials Chemistry A, 2015; 104 citations. [corresponding]

Surface-Functionalized Silicon Nanoparticles as Anode Material for Lithium-Ion Battery — ACS Applied Materials & Interfaces, 2018; 102 citations.

Understanding of pre-lithiation of poly(acrylic acid) binder: Striking the balances between the cycling performance and slurry stability for silicon-graphite composite electrodes in Li-ion batteries — Journal of Power Sources, 2019; 82 citations.

Book Chapters

Electrolytes used in silicon anodes — in Lithium-ion Batteries Enabled by Silicon Anodes, Institution of Engineering and Technology (IET), 2021; ISBN 9781785619557.

Redox Shuttle Additives for Lithium-Ion Battery — in Lithium Ion Batteries — New Developments, InTechOpen, 2012; ISBN 978-953-307-900-4.

US Patents (selected, 25 total)

US 12,322,757 — Isatin derivative redoxamer for electrochemical device (2025). Inventors: Bheemireddy, S.R.; Zhang, L.; Zhang, Z..

US 12,015,156 — Solvents and slurries comprising a poly(carboxylic acid) binder for silicon electrode manufacture (2024). Inventors: Zhang, L.; Shi, Z.; Jansen, A.; Trask, S..

US 11,715,844 — Isatin derivative redoxmer for electrochemical device (2023). Inventors: Bheemireddy, S.R.; Zhang, L.; Zhang, Z..

US 11,271,237 — Organic redox molecules for flow batteries (2022). Inventors: Zhang, L.; Zhang, J.; Shkrob, I.A..

US 11,532,818 — Solvents and slurries comprising a poly(carboxylic acid) binder for silicon electrode manufacture (2022). Inventors: Zhang, L.; Shi, Z.; Jansen, A.; Trask, S..

US 11,411,221 — Binders for silicon electrodes in lithium-ion batteries (2022). Inventors: Shi, Z.; Zhang, L..

US 10,840,531 — Two-electron redox active molecules with high capacity and energy density for energy storage (2020). Inventors: Huang, J.; Zhang, L.; Burrell, A.K.; Zhang, Z..

US 10,535,891 — Two-electron redox catholyte for redox flow batteries (2020). Inventors: Zhang, J.; Zhang, L.; Shkrob, I.A..

US 10,637,099 — Annulated tetra-substituted hydroquinone ether-based redox shuttle additives (2020). Inventors: Zhang, L.; Zhang, J.; Shkrob, I.A.; Zhang, Z..

US 10,622,676 — Functional organic salt for lithium-ion batteries (2020). Inventors: Zhang, L.; Zhang, J.; Shkrob, I.A..

Funded Research Projects (selected)

Advanced Electrolyte Additives for PHEV/EV Lithium-ion Battery — 2010 – 2014
DOE ABR (Applied Battery Research) · Researcher → Co-investigator

Organic Molecular Engineering for Non-Aqueous Redox Flow Batteries (NRFBs / ROMs) — 2013 – 2020
JCESR — DOE BES Energy Frontier Research Center · Theme Lead · Sprint team leader · Y3 milestone lead

Silicon Consortium Project — Science of Manufacturing for Silicon Anodes 2014 – 2021
DOE EERE / Vehicle Technologies Office · Principal Investigator (binders sub-thrust)

Flow Batteries Technology Strategy Assessment / Storage Innovations 2030 2022 – 2023
DOE — long-duration storage roadmap · Co-author · DOE flow-battery information lead

High-Density Energy Storage in Solvent-Free Redox Flow Batteries for Flexible Electric Grid 2018 – 2020
Argonne LDRD (2018-100) · Principal Investigator (with Ilya A. Shkrob)

Hybrid Membranes for Flow Batteries 2021 – 2023
Argonne LDRD (2021-0087) · Project member · membrane sub-thrust (with Jeffrey Elam)

Invited Talks (top 8)

Invited speaker (title not yet captured) — MRS Spring Meeting — ES05, Seattle, WA, 2024.

Invited speaker (title not yet captured) — ECS 241st Meeting — Lithium Ion Batteries, Vancouver, Canada, 2022.

A Data-Driven Approach to Probe the Stability of Dialkoxy Benzene Catholytes for Nonaqueous Redox Flow Batteries — International Battery Seminar — Battery Mgmt Systems, Online, 2021.

Invited speaker (title not yet captured) — ECS 235th Meeting — Large-Scale Energy Storage 10, Dallas, TX, 2019.

Stable Dialkoxybenzenes as Catholyte Materials for Non-Aqueous Redox Flow Batteries — MRS Spring Meeting, Phoenix, AZ, 2018.

Stable Annulated Dialkoxybenzenes as Catholyte Materials for Non-Aqueous Redox Flow Batteries — International Battery Seminar & Exhibit, Fort Lauderdale, FL, 2018.

Recent Advances in Organic Redox-Active Materials for Flow Batteries — ICESI Annual Meeting, Dalian, China, 2018.

Redox Shuttle Additives towards Safer Lithium-Ion Batteries — International Battery Seminar & Exhibit, Fort Lauderdale, FL, 2017.

Service & Peer Review

- **Co-author, DOE Flow Batteries Technology Strategy Assessment** (2023) — long-duration storage roadmap.
- Storage Innovations 2030 contributor (DOE framework, 2022).
- JCESR Theme / Sprint team leadership (3 universities + 2 national labs, 2015 – 2017).
- JCESR 2.0 self-reporting redoxmer Y3 milestone leader (2020).
- **DOE SBIR Reviewer** (2013 – 2022, 10 years).
- **NSF Review Panel for Energy Systems** (2019).
- **DOE Energy Storage Innovations Prize Reviewer** (2023).
- Reviewer for 10+ international journals: J. Power Sources, ACS Energy Lett., Chem. Commun., Adv. Energy Mater., Nano Energy, J. Mater. Chem. A, Chemistry of Materials, ChemSusChem, Electrochim. Acta, ACS Appl. Energy Mater.

Mentorship

Supervised **6 postdoctoral fellows + 3 graduate students** at ANL (2013 – 2023). Direct mentees who first-authored papers under my supervision: Jinhua Huang, Jingjing Zhang, Lily A. Robertson, Bin Hu. Other group members include Sisi Jiang, Zhangxing Shi, Sambasiva R. Bheemireddy, Wentao Duan, Hossam Farag, Aman Preet Kaur. Cross-institution Sprint teams spanned MIT, PNNL, UIUC, UMichigan, Purdue.